

TUTORIAL SEVEN

Real-Time OS and Virtualization

1. Indicate whether the following statements are true or false. Justify your answers.
 - a) Real-time CPS applications usually require a response within a very short time.
 - b) A guest OS abstracts H/W and provides an interface between H/W and hypervisor.
 - c) Managing the interactions between a guest OS and its applications is one of the key functions of a hypervisor.
 - d) Under a Type-1 para-virtualized hypervisor, the hypervisor directly interacts with the H/W and provides a virtual call API to interact with the guest OS.
2. You are tasked with setting up a virtualized environment to run a variety of workloads, including data processing (heavyweight) applications and web servers (lightweight), on a number of physical servers. The goal is to maximize performance and resource utilization while ensuring flexibility across different workloads.
3. Consider the following set of periodic real-time processes:

$P_1<5, 2, 5>$

$P_2<10, 3, 10>$

$P_3<20, 5, 20>$

Construct **all possible** schedules for this process set under the earliest deadline first (EDF) scheduling strategy. Is this process set **schedulable** under EDF?